

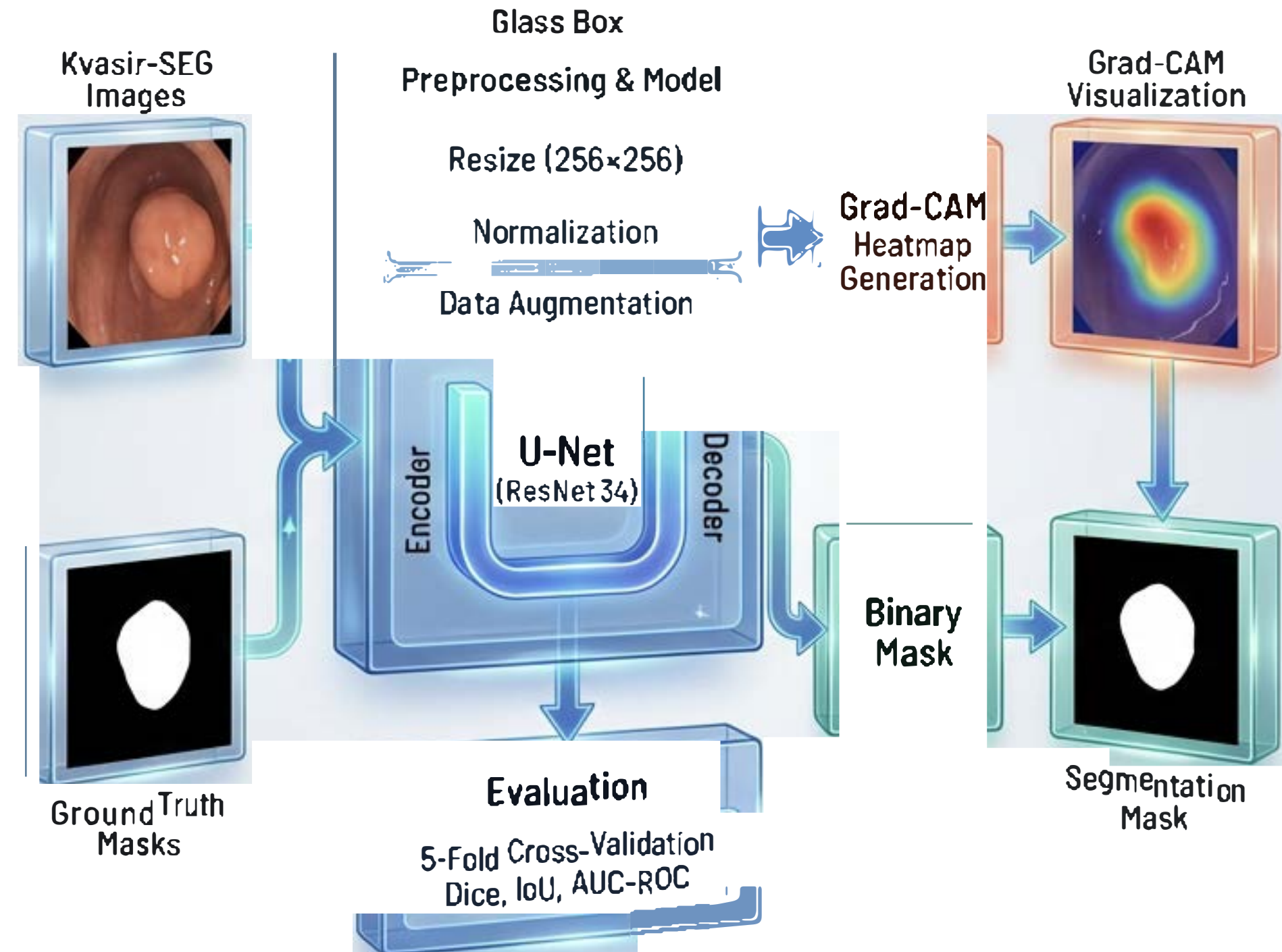


NORTHWESTERN
UNIVERSITY.

Improved Segmentation of Polyps and Visual Explainability Analysis

A Deep Learning Framework Integrating U-Net, ResNet-34, and Gradient-Weighted Class Activation Mapping

Akwasi Asare (Ghana Communication Technology University)
Thanh-Huy Nguyen (Carnegie Mellon University)
Ulas Bagci (Northwestern University)



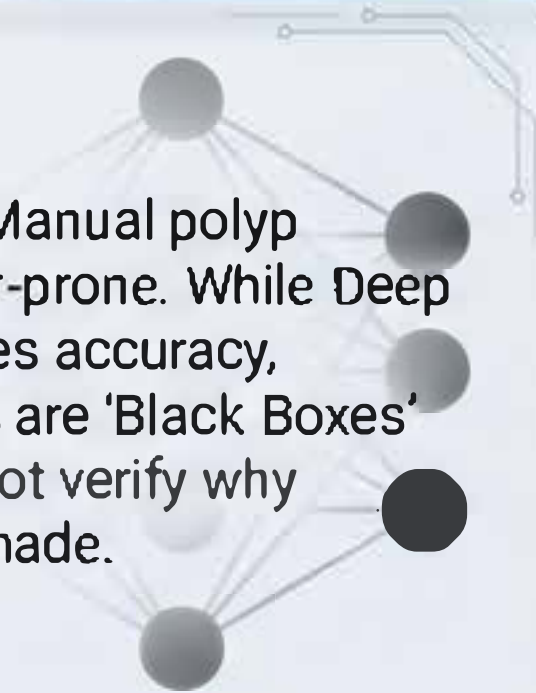
Executive Summary: Bridging the Gap Between AI Accuracy and Clinical Trust

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THE CHALLENGE



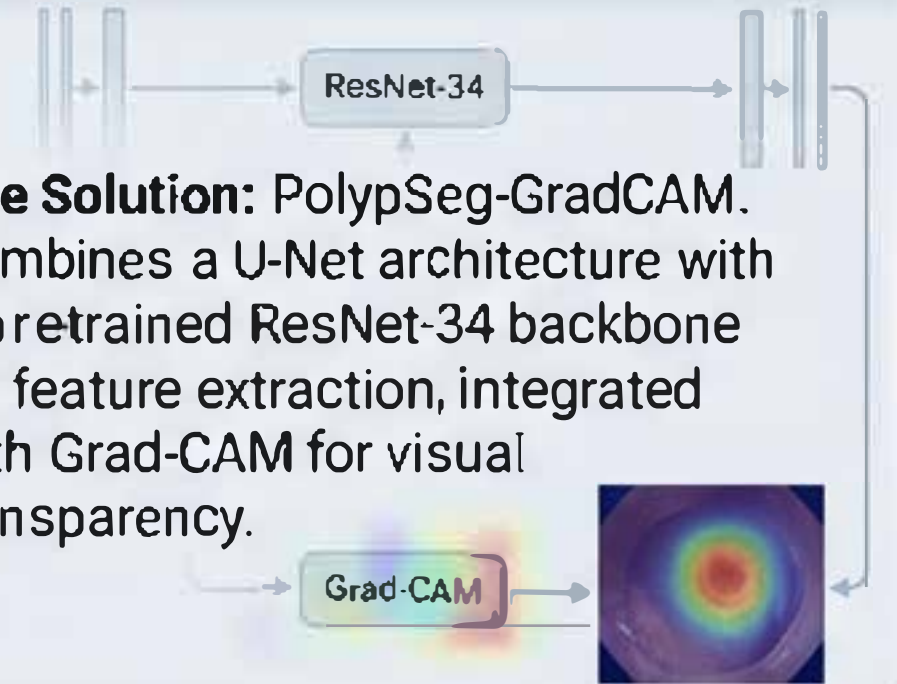
The Challenge: Manual polyp detection is error-prone. While Deep Learning improves accuracy, standard models are 'Black Boxes'—clinicians cannot verify why a decision was made.



THE INNOVATION



The Solution: PolypSeg-GradCAM. Combines a U-Net architecture with a pretrained ResNet-34 backbone for feature extraction, integrated with Grad-CAM for visual transparency.



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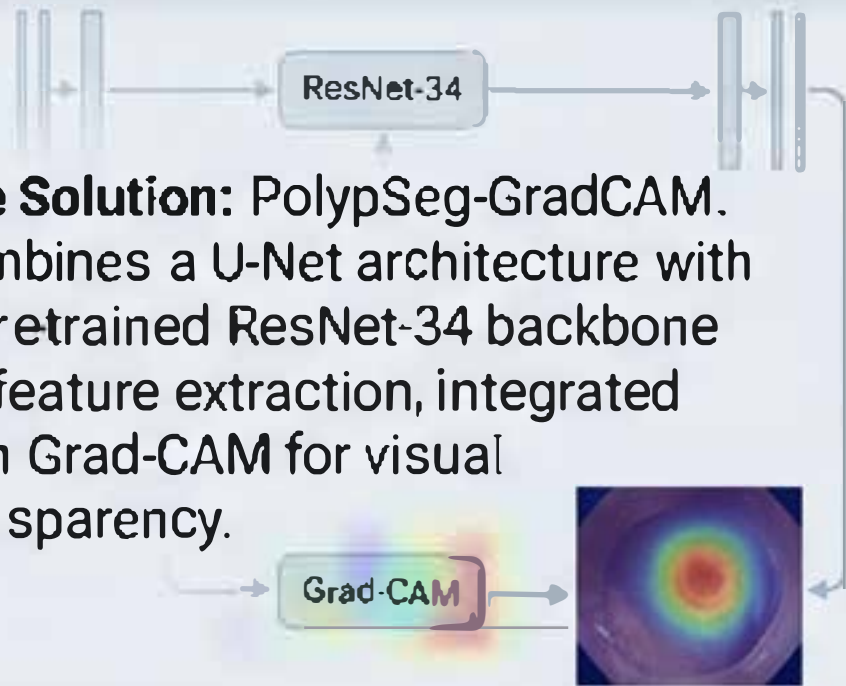


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THE INNOVATION



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KEY METRICS

ACCURACY

0.8902



Mean Dice Coefficient

RELIABILITY

5-Fold



CrossValidation

SPEED

176.54



FPS (Realtime)

INTERPRETABILITY

Heatmap



Visual Verification

The Global Burden of Colorectal Cancer

Key Data

1.9 Million

New cases annually (2020)

**2nd Leading
Cause**

Of cancer-related deaths
worldwide

>930,000

Fatalities per year

+63%

Projected increase in
cases by 2040

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New cases annually (2020)

2nd Leading Cause

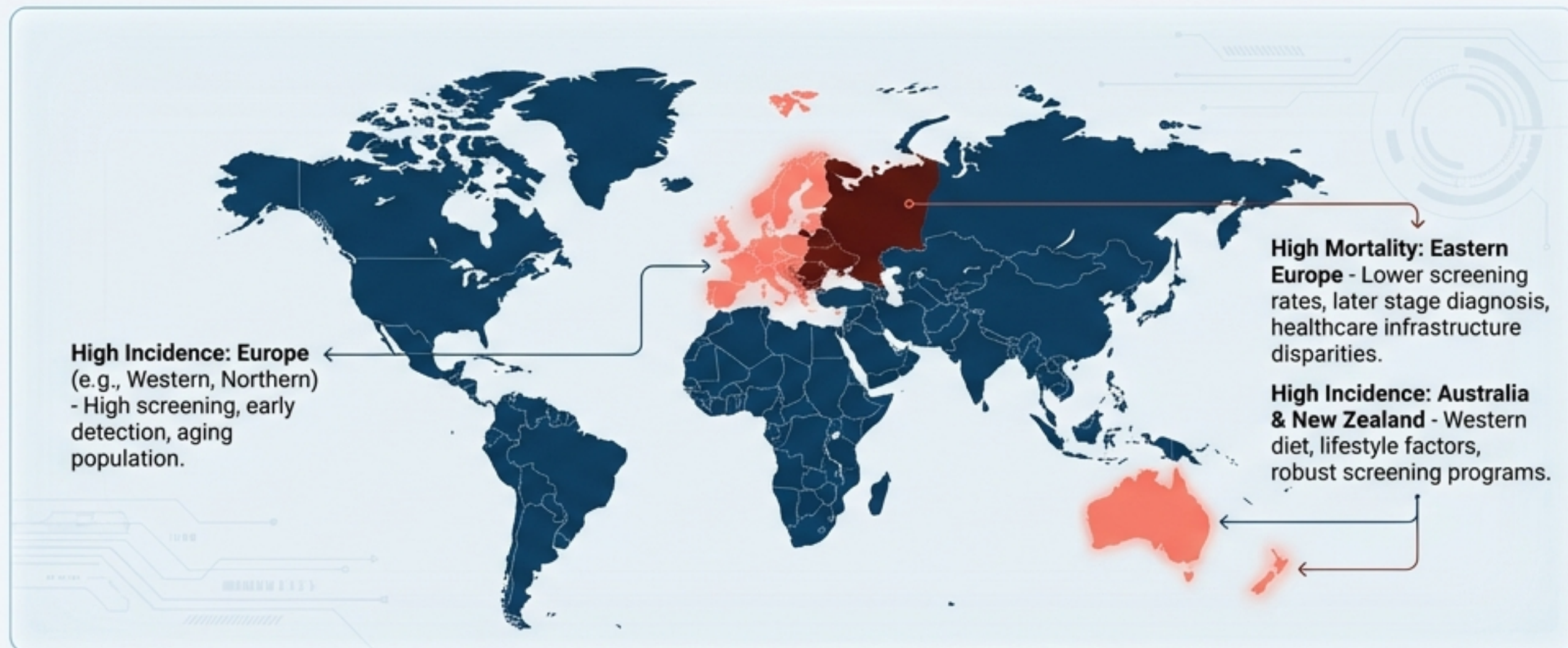
Of cancer-related deaths worldwide

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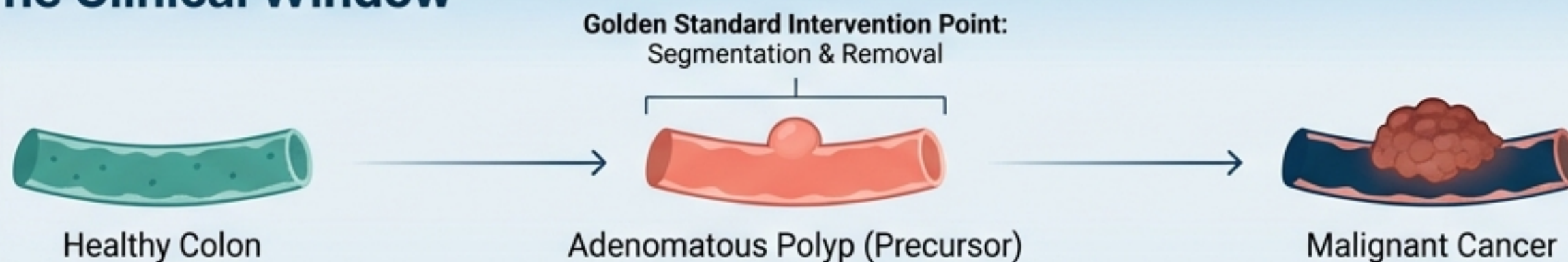
Fatalities per year

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Projected increase in cases by 2040

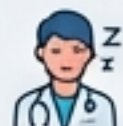


The Clinical Window



Inter The Diagnostics Dilemma

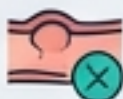
Human Limitations



Observer Variability: Fatigue and experience levels affect detection.



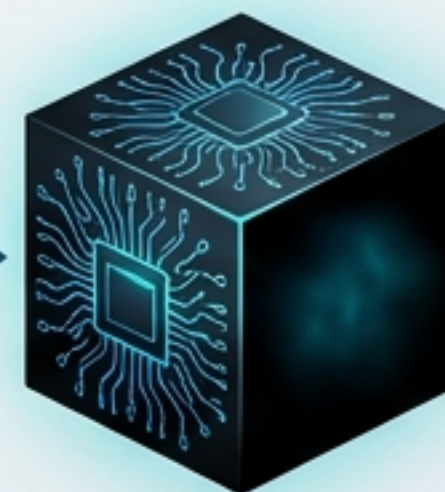
Difficult Imaging: Specular highlights, stool residues, and bubbles obscure view.



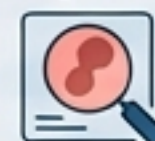
Miss Rate: Subtle flat polyps are often overlooked.

The AI Trust Gap

Image



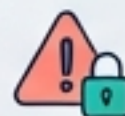
Diagnosis



Deep Learning (CNNs): High automation potential.



The Problem: Lack of interpretability.

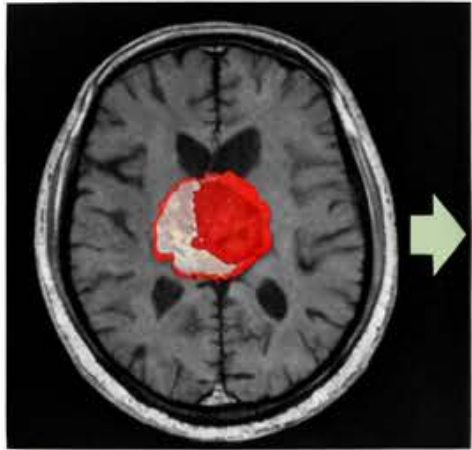


The Risk: In high-stakes oncology, a 'Black Box' decision is insufficient for clinical trust.

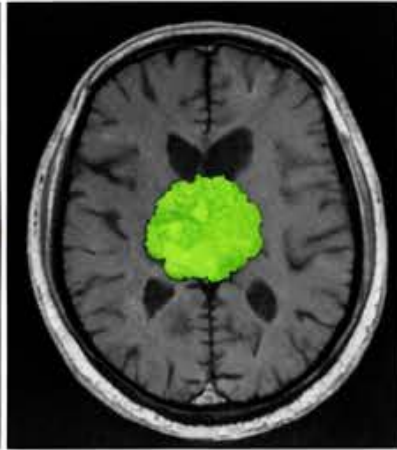
“ Clinicians require not only high accuracy but also interpretability to trust automated decisions.

Scenario A: Correct for Correct Reasons

Input MRI & Ground Truth



Segmentation Output



TRUSTWORTHY MODEL

- ✓ Aligned with Pathology
- ✓ Clinically Sound Reasoning
- ✓ High Clinical Trust

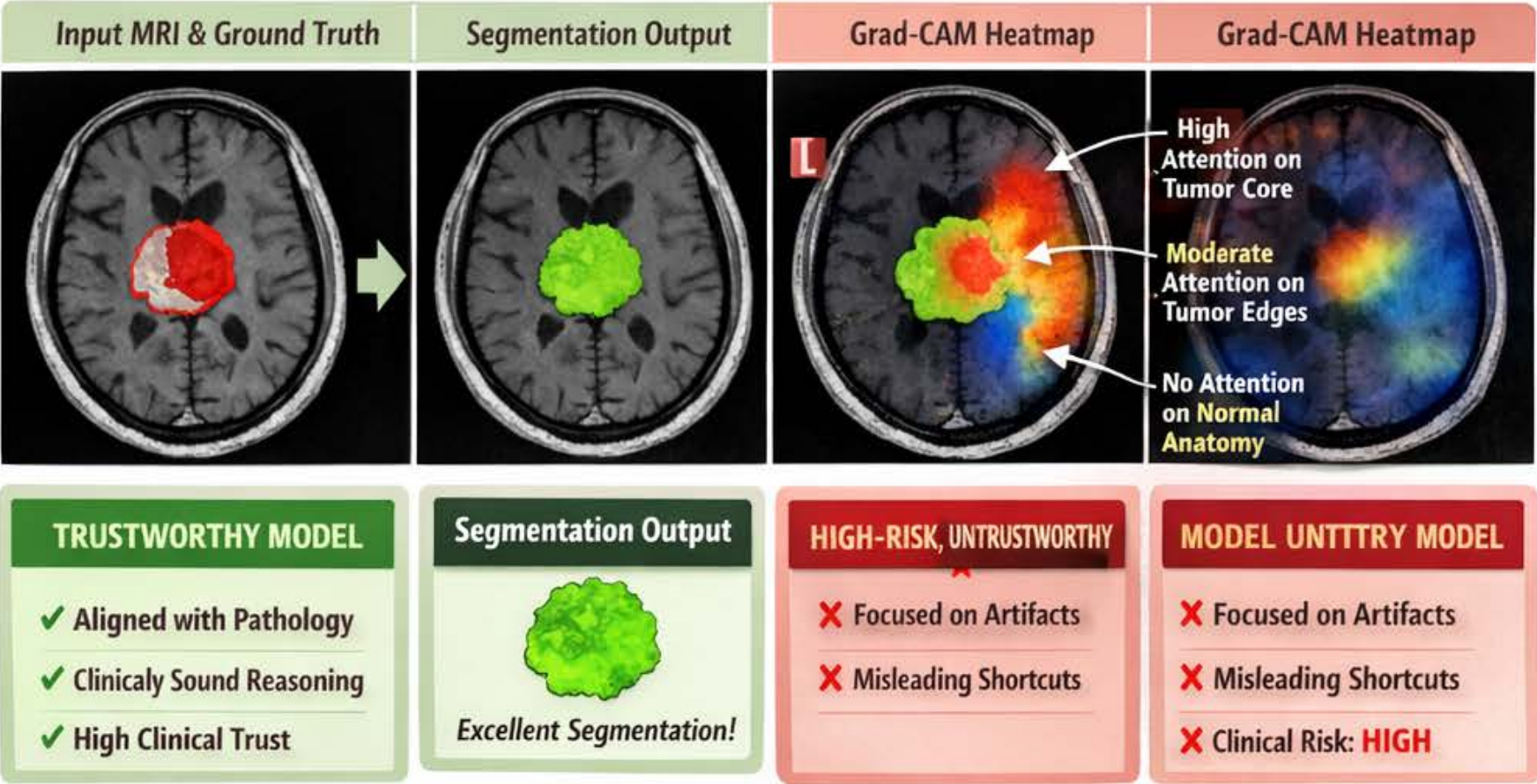
Segmentation Output



Excellent Segmentation!

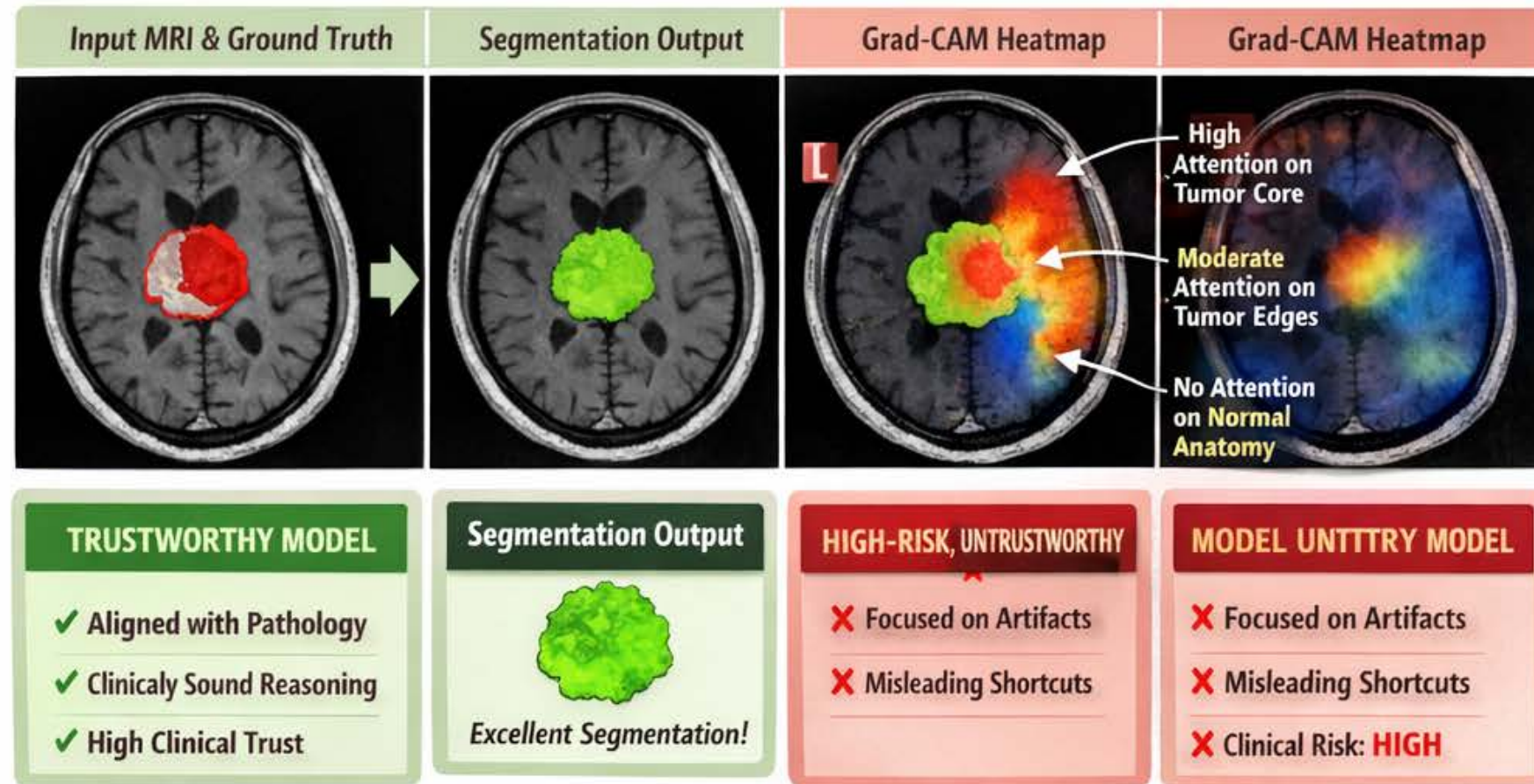
Scenario A: Correct for Correct Reasons

Scenario B: Correct for Wrong Reasons

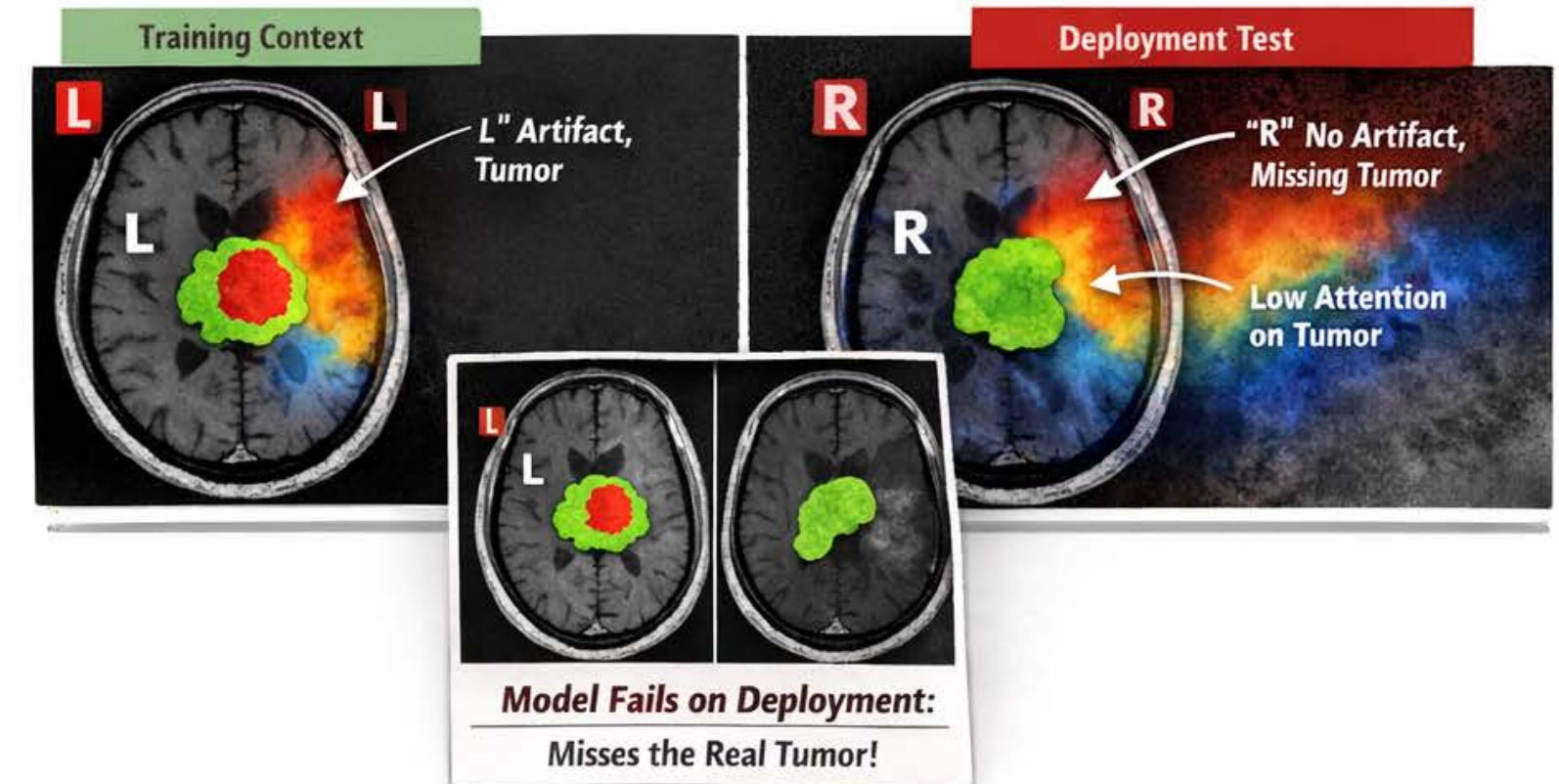


Scenario A: Correct for Correct Reasons

Scenario B: Correct for Wrong Reasons

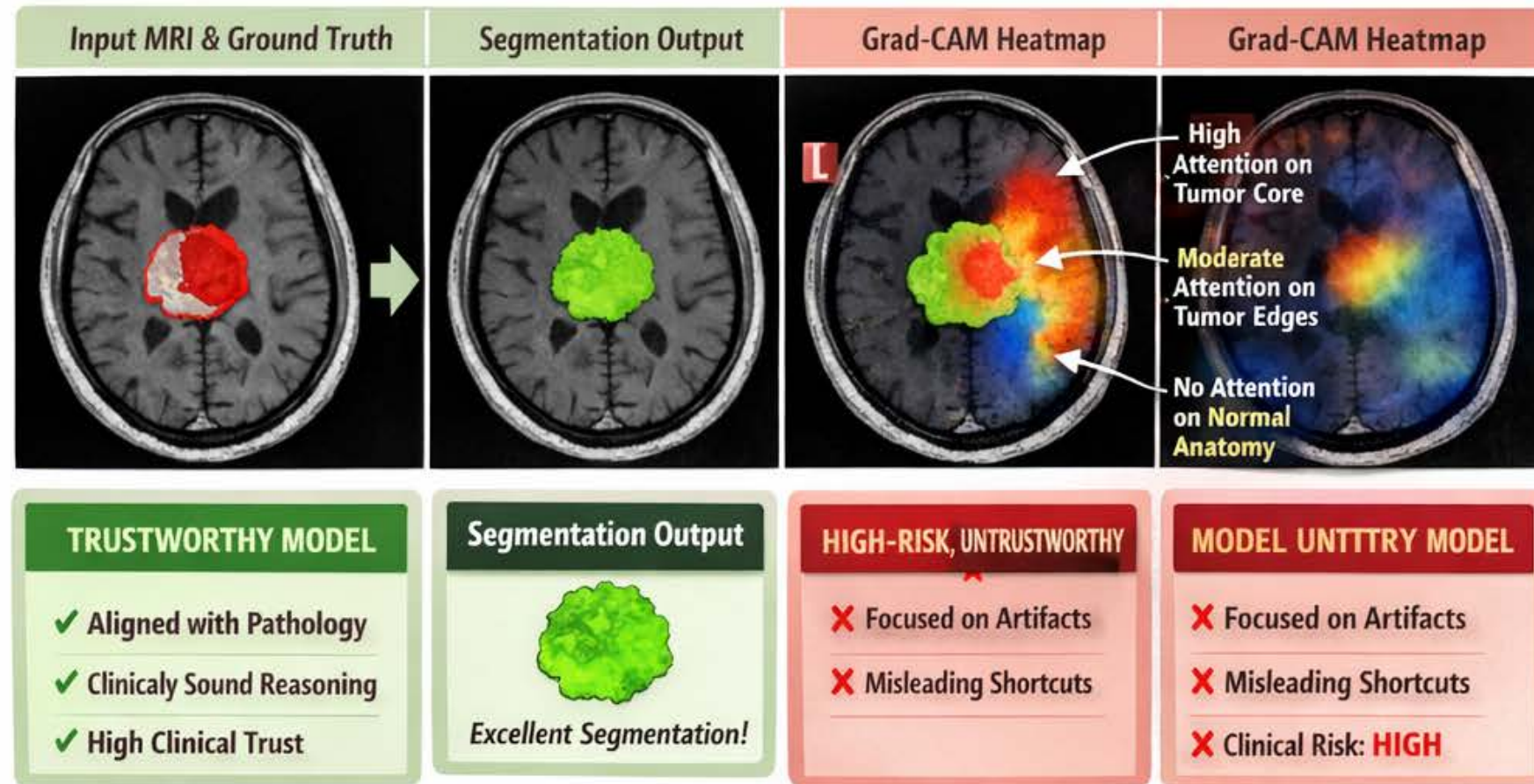


The Fatal Flaw Revealed

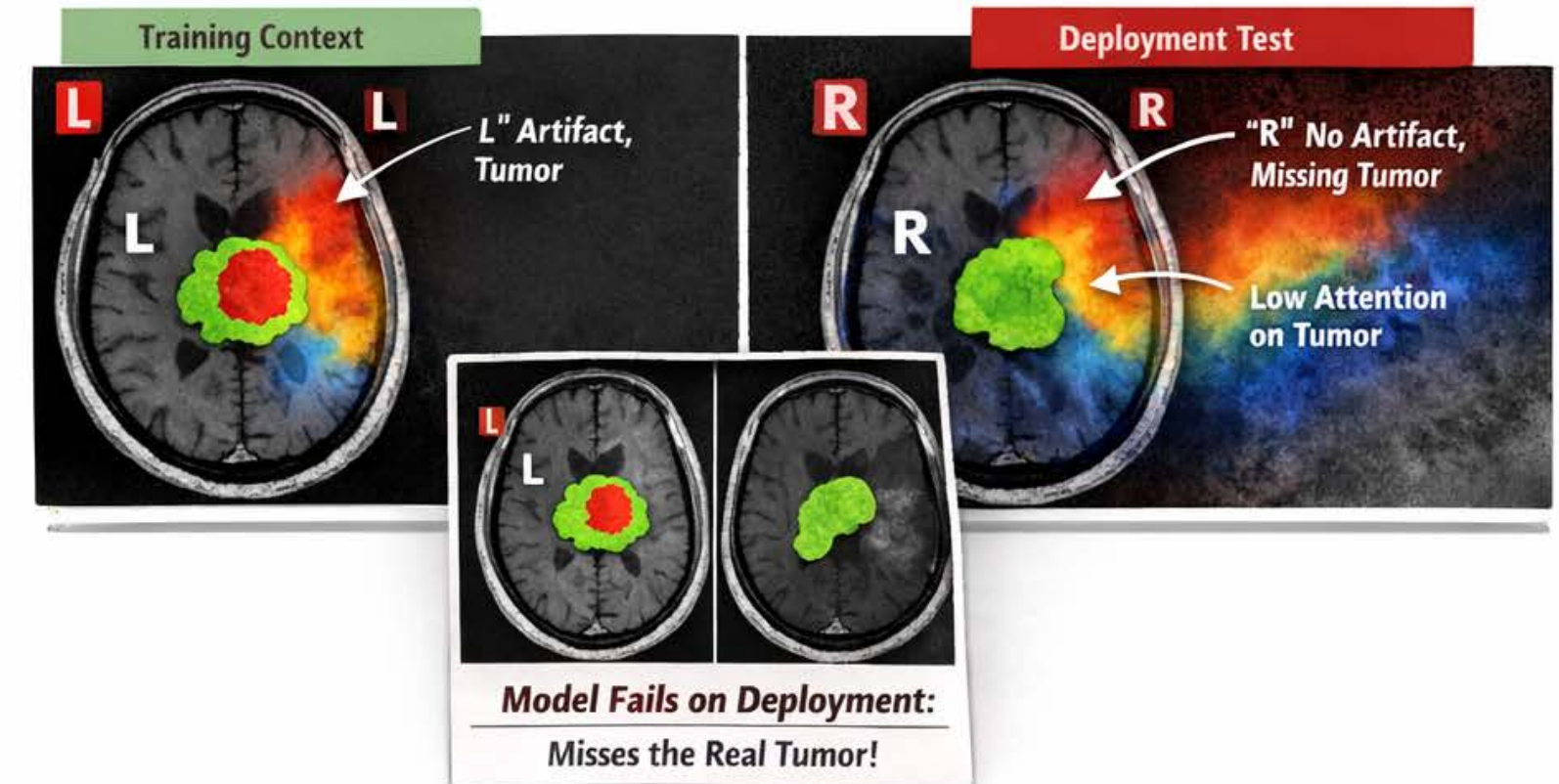


Scenario A: Correct for Correct Reasons

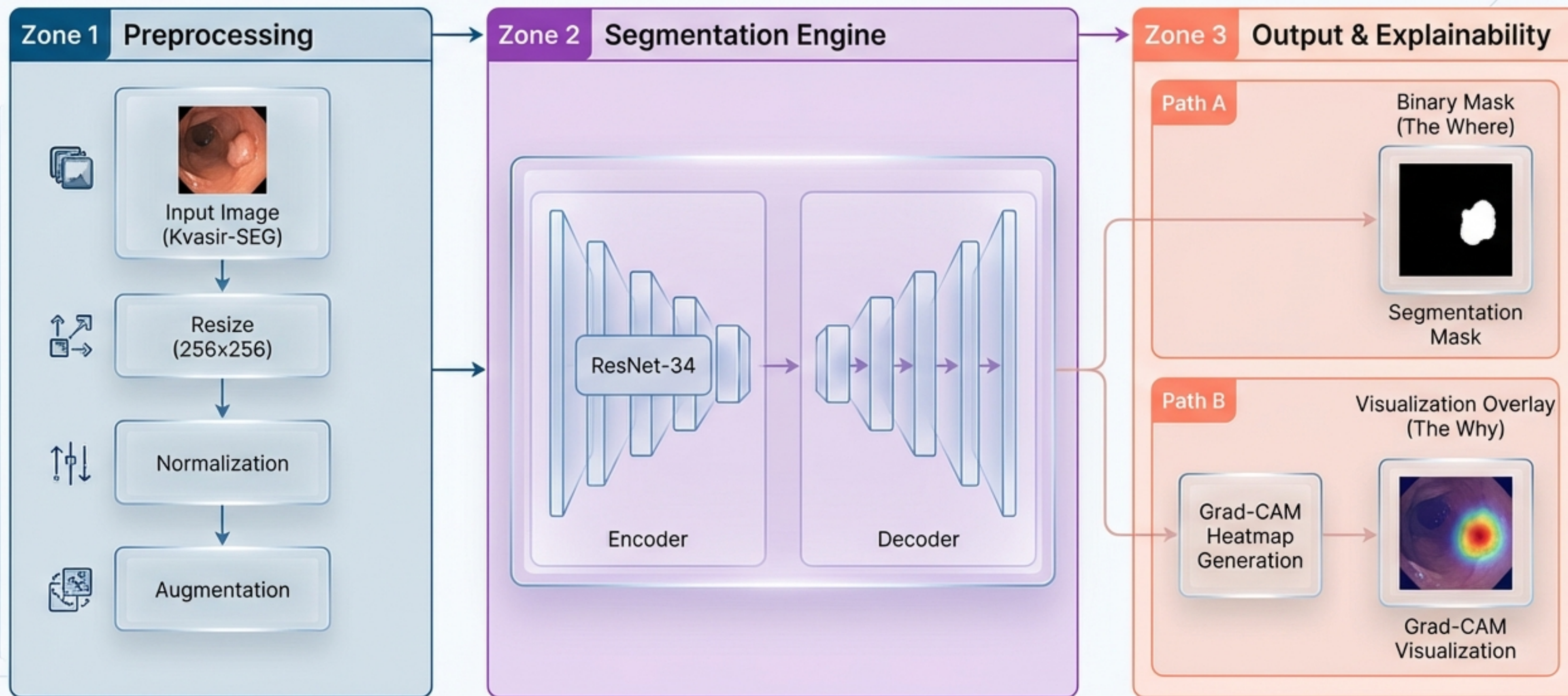
Scenario B: Correct for Wrong Reasons



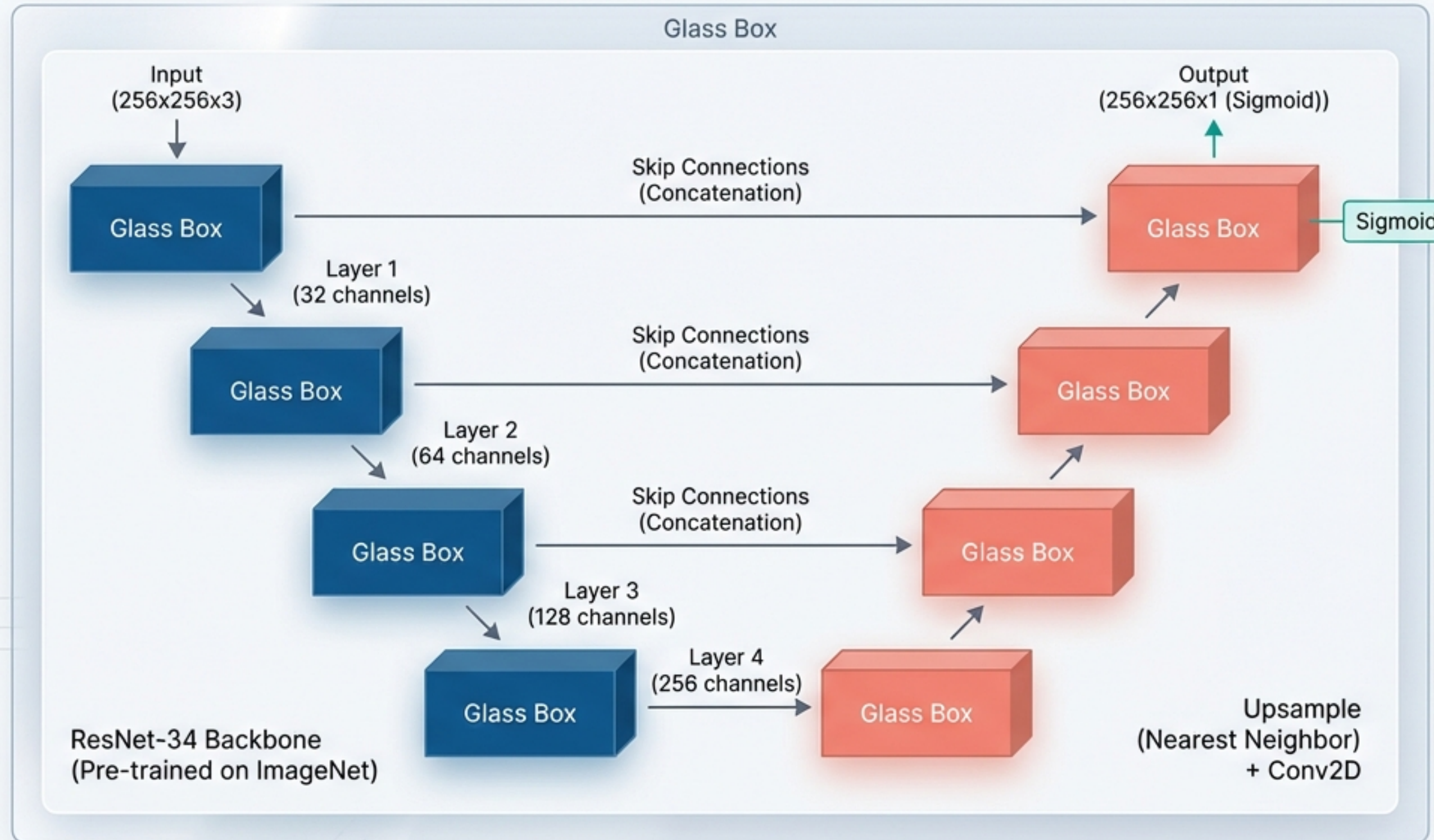
The Fatal Flaw Revealed



The PolypSeg-GradCAM Framework



Architecture: U-Net Enhanced with ResNet-34



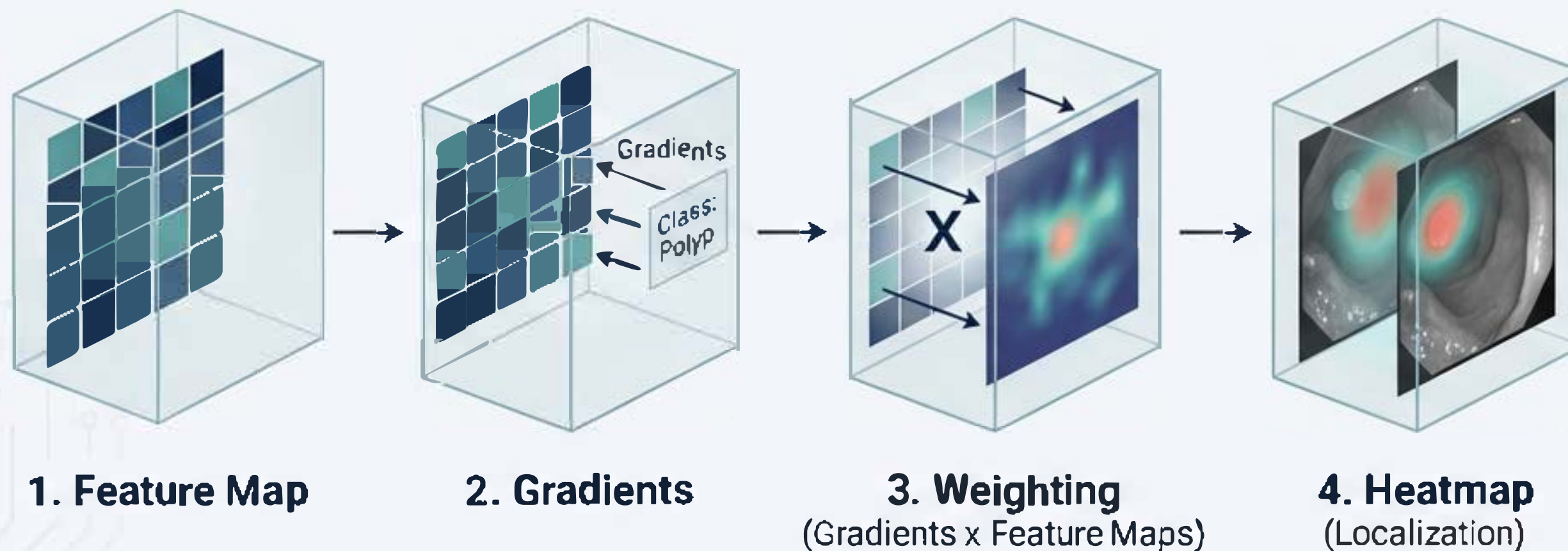
Why ResNet-34?

Replaces standard encoder with ImageNet pre-trained weights.

Accelerates convergence and improves feature extraction on limited medical datasets.

JetBrains Mono
Trainable Parameters:
24.44 Million

Opening the Black Box: Grad-CAM



Technique:

Grad-CAM computes the gradients of the target class with respect to the final decoder feature maps.

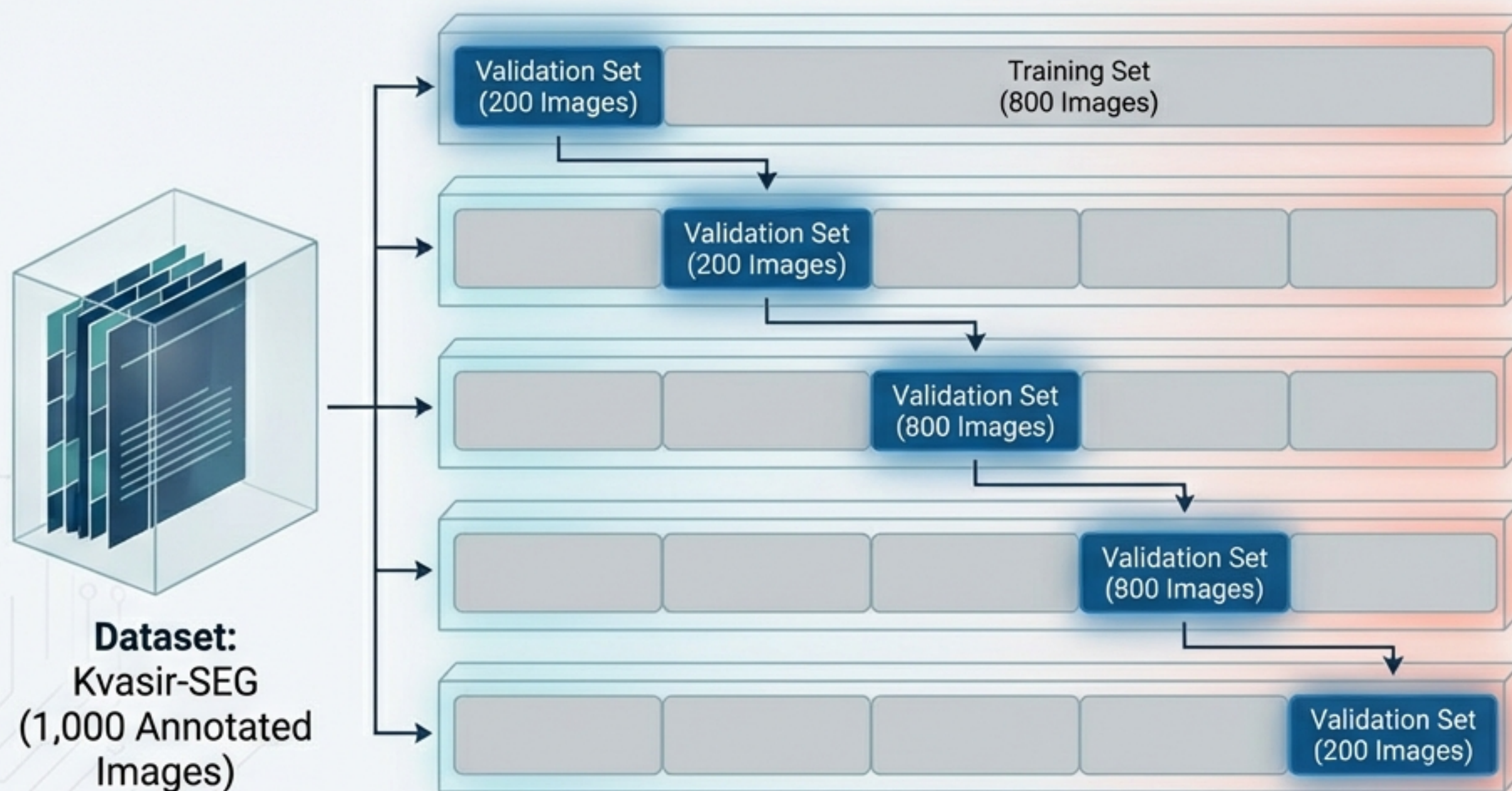
Localization:

Generates channel-wise importance weights to create a localization map.

Differentiation:

Crucial Value: Ensures the model is segmenting based on the polyp tissue, not background artifacts like bubbles or surgical tools.

Experimental Rigor: 5-Fold Cross-Validation



Dataset:
Kvasir-SEG
(1,000 Annotated
Images)

Methodology: Data partitioned into 5 disjoint subsets.
Every image evaluated exactly once as a test sample.

Purpose:

Mitigates selection bias
and proves robustness
against "lucky splits".

Preprocessing:

Z-score normalization
(ImageNet stats) &
Albumentations
(Flips/Rotations).

Total Folds: 5 |
Iterations: 5 |
Total Evaluations:
1000 Images

Quantitative Results: Segmentation Performance

Mean Dice Coefficient

0.8902 ± 0.0125 (SD)

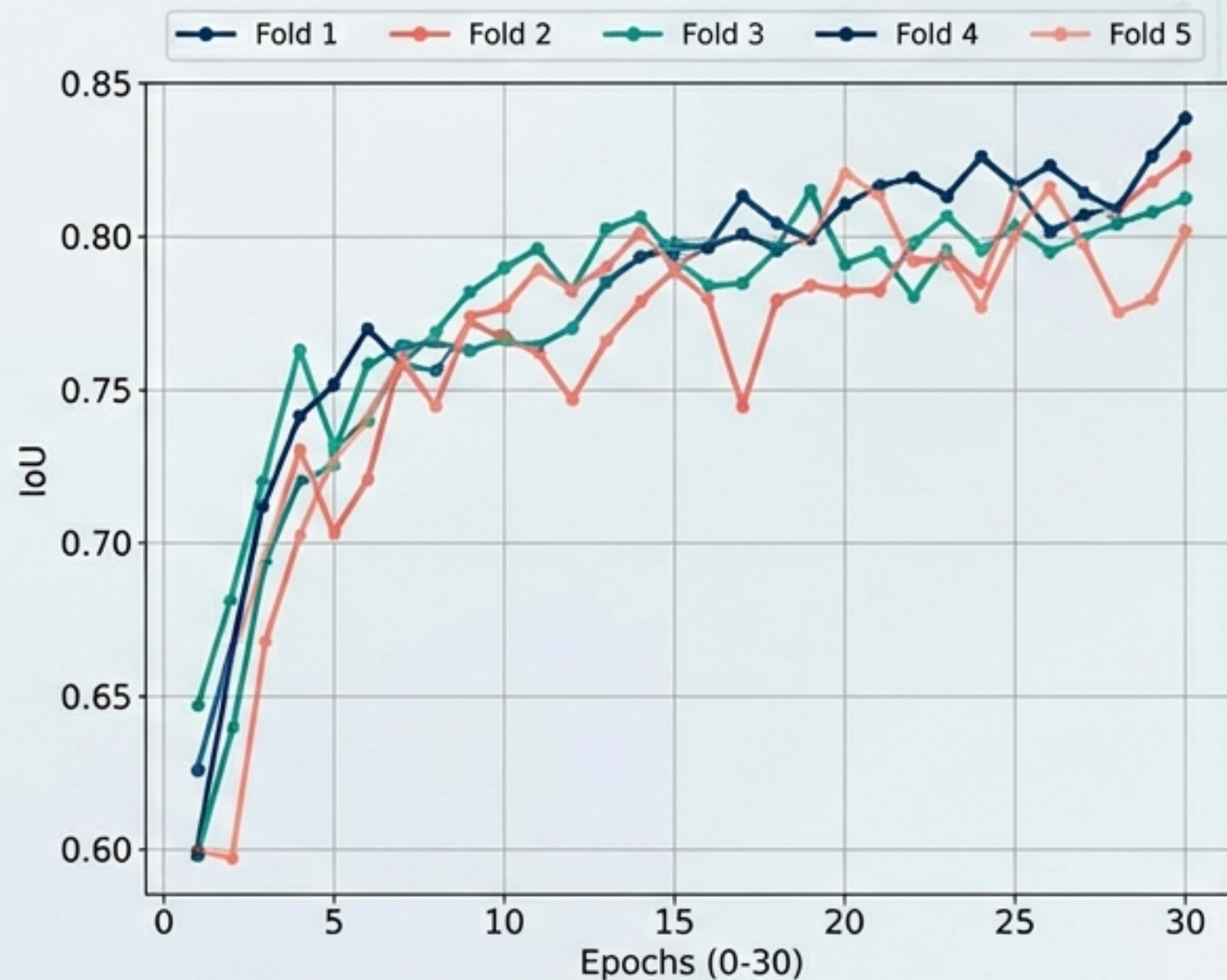
Mean IoU

0.8023

Sensitivity

0.9058

IoU Across Folds



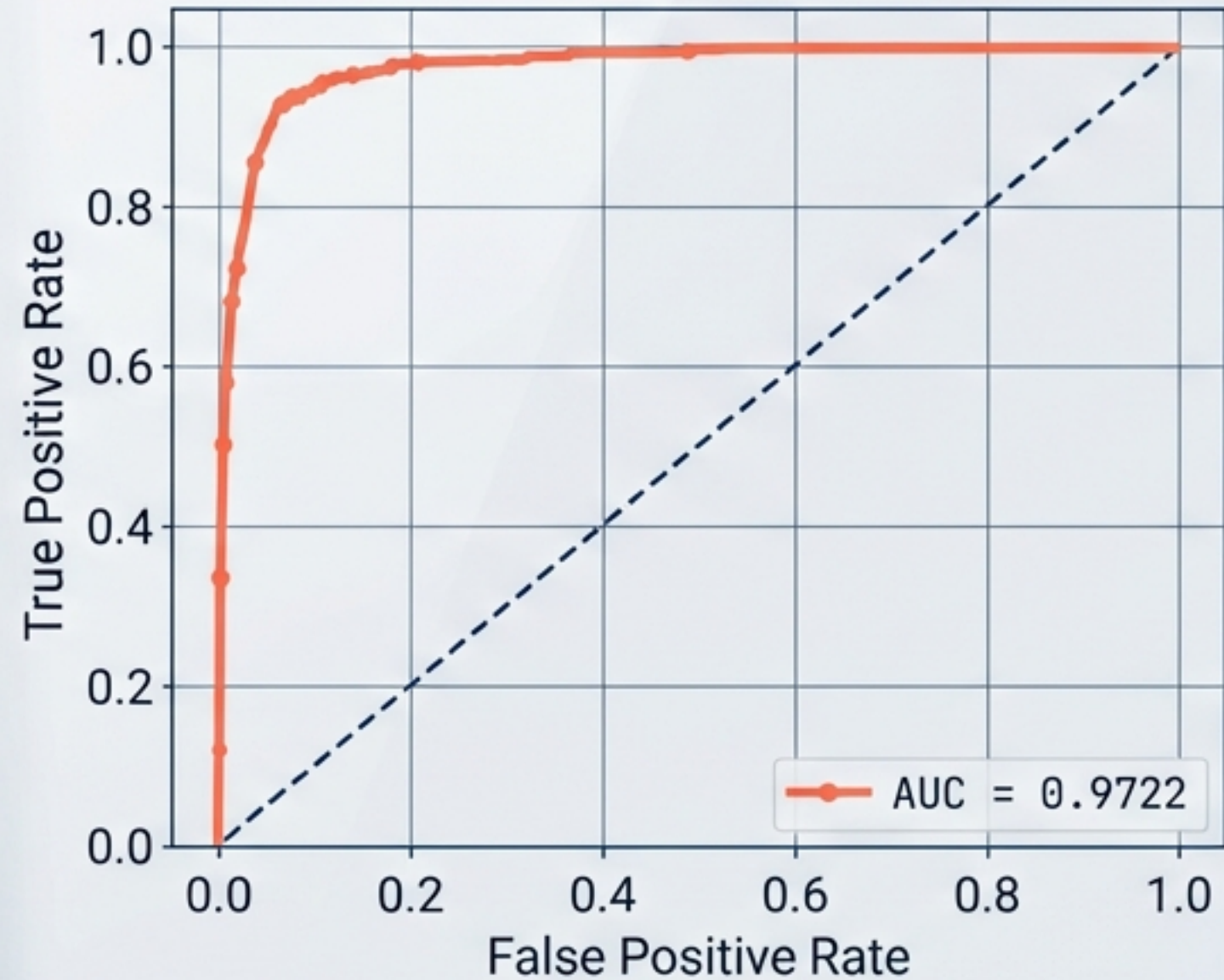
Benchmarking Against State-of-the-Art

Model	Dice Coefficient	IoU
ResUNet (Baseline)	0.7878	0.7778
ResUNet++	0.8198	0.7318
TransAttUnet	0.8864	0.8020
CaraNet	0.8923	0.8218
PolypSeg-GradCAM (Ours)	0.8902	0.8023

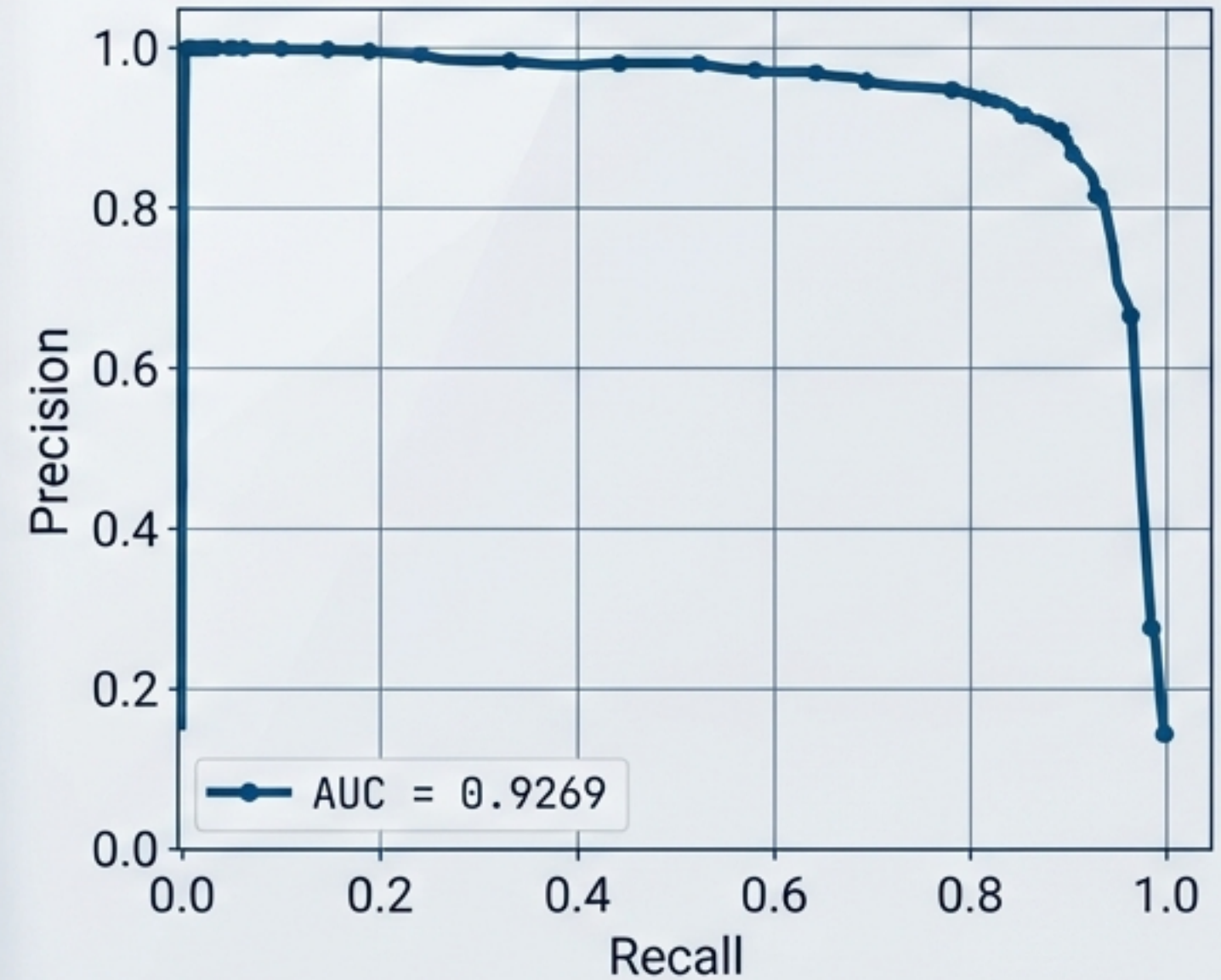
Competitive performance with complex architectures (CaraNet, PraNet) while uniquely prioritizing visual explainability.

Discriminative Ability & Separability

ROC Curve



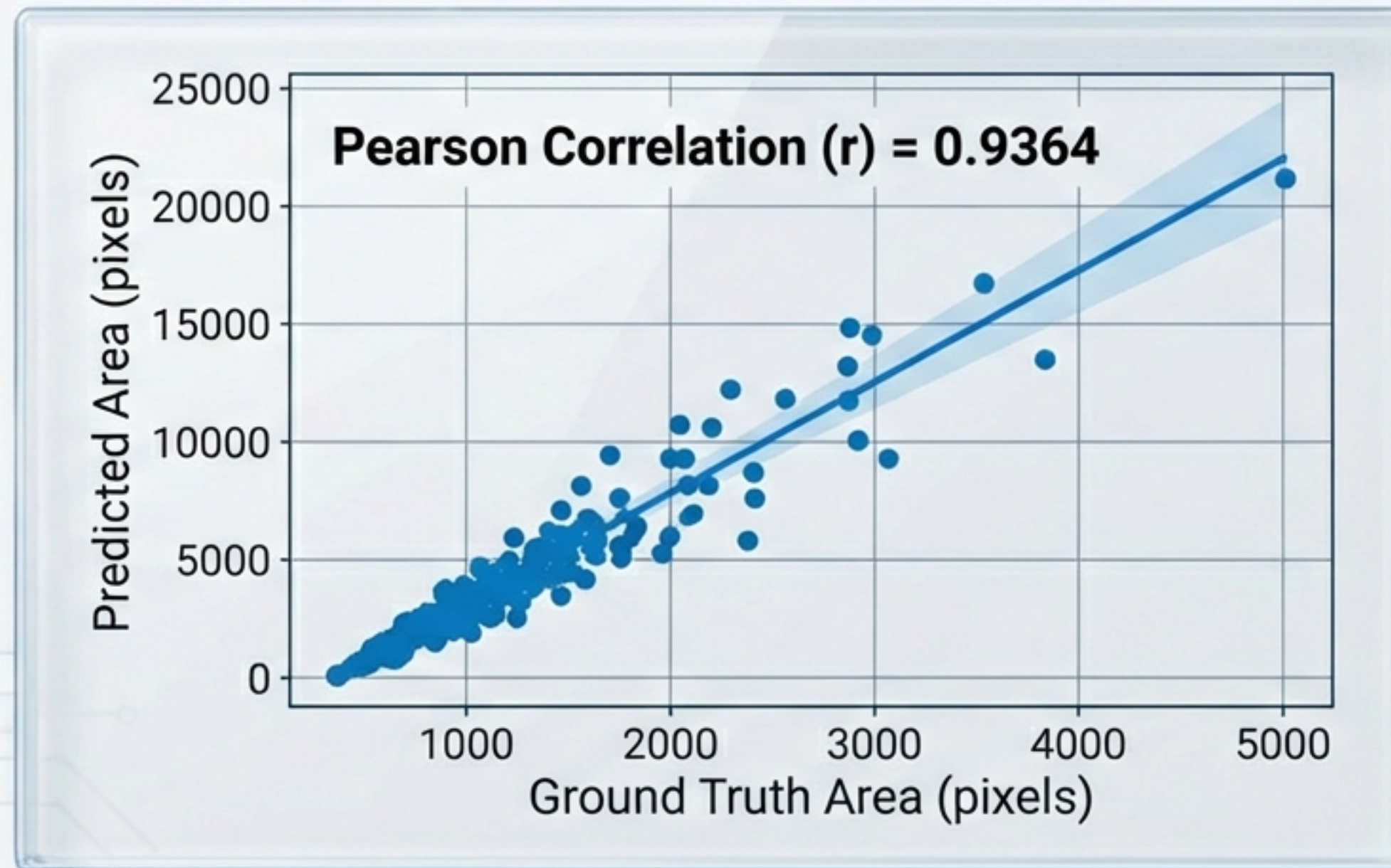
Precision-Recall Curve



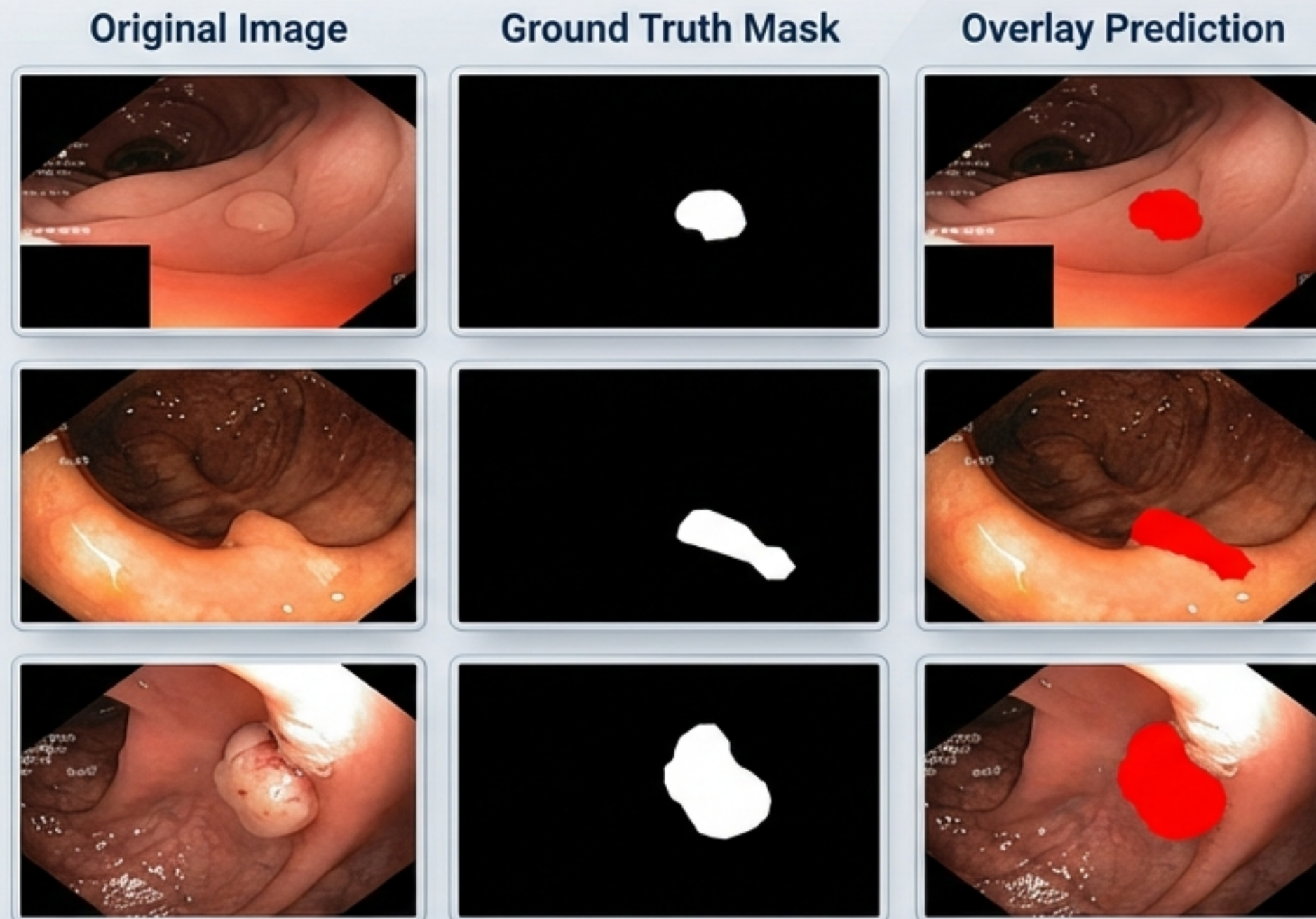
Optimal Threshold: 0.5500. **Specificity:** 98.24%. Maximizes ability to distinguish polyp tissue from background.

Clinical Utility: Automated Polyp Sizing

Accurate size estimation is critical for malignancy risk stratification. The model demonstrates high correlation with expert annotations.



Qualitative Results: Segmentation Fidelity



Visual comparison of original polyp images, ground truth annotations, and model segmentation predictions. The model demonstrates high fidelity and precise boundary adherence, accurately segmenting polyps of varying shapes and under challenging lighting conditions.

Visualizing Trust: Grad-CAM Heatmaps

Original Input



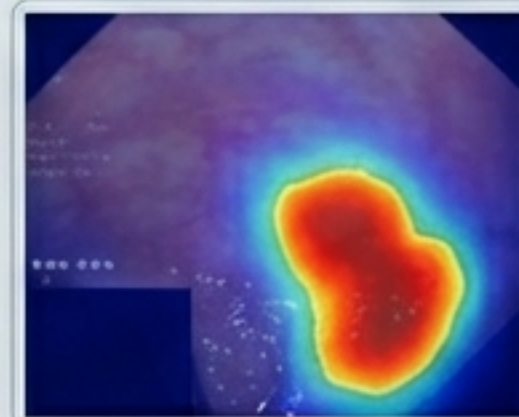
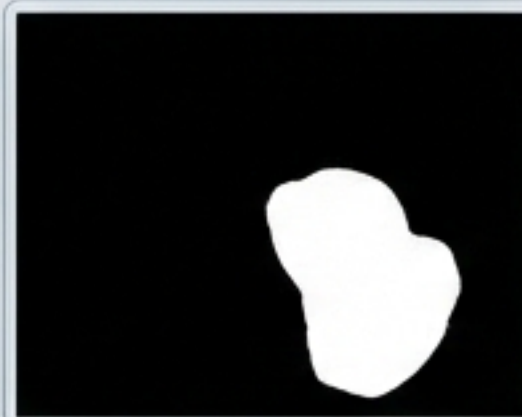
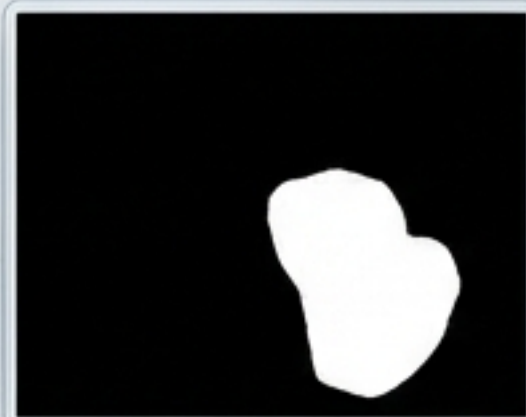
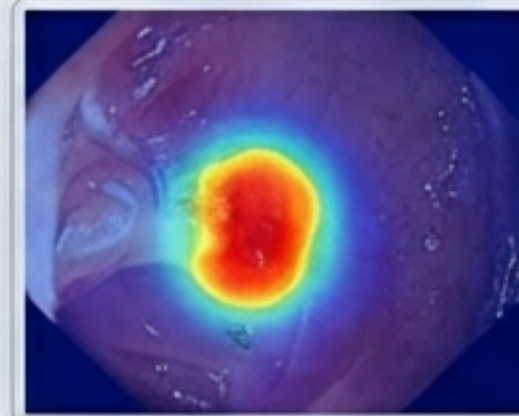
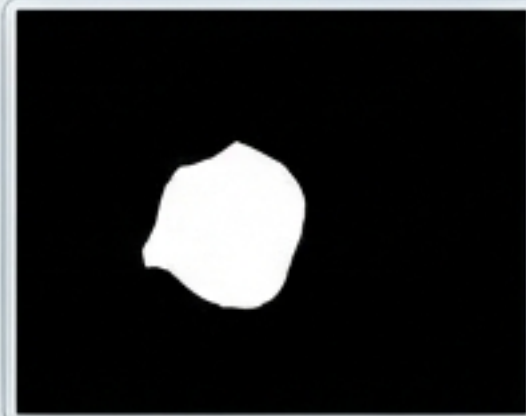
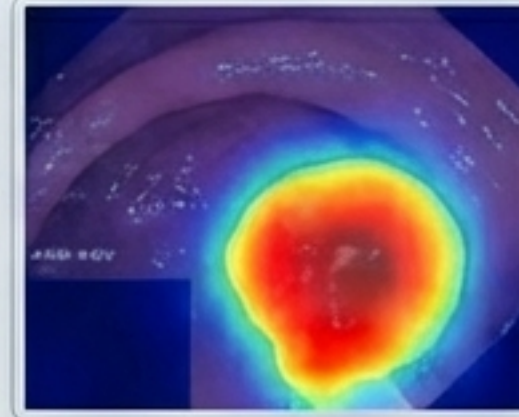
Ground Truth



Prediction



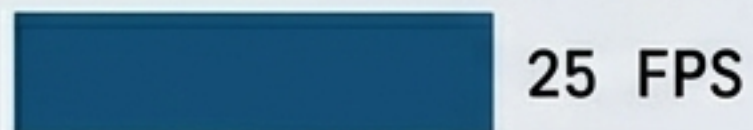
Grad-CAM Explainability



Model focuses strictly on polyp tissue, ignoring mucosal folds and artifacts.

Computational Efficiency & Real-Time Readiness

Real-Time Requirement



PolypSeg-GradCAM



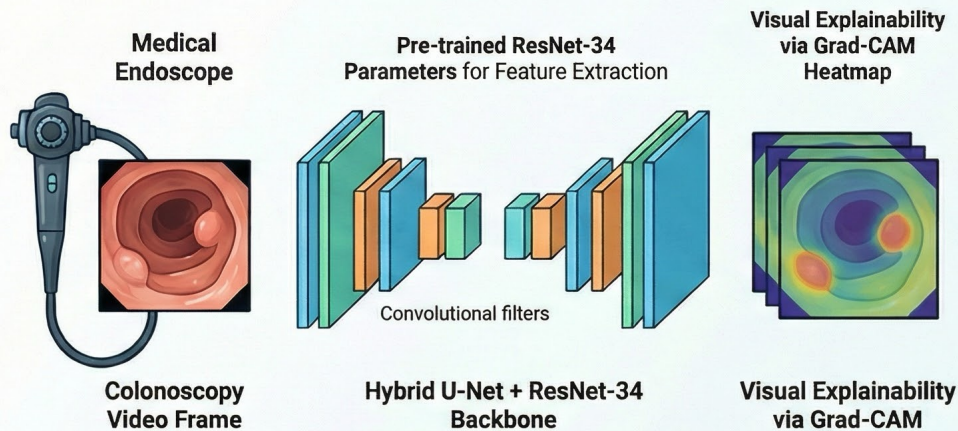
Inference Speed	Latency	Complexity
176.54 FPS	5.66 ms / image	15.70 GFL0Ps

The system is ~7x faster than the real-time requirement, making it suitable for live video integration in endoscopy suites.

SUMMARY

PolypSeg-GradCAM is an explainable deep learning framework for automated polyp detection during colonoscopy, combining high-performance AI with Grad-CAM heatmaps for real-time visual transparency.

The Explainable Segmentation Pipeline



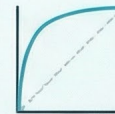
176.54 Frames Per Second

High-speed inference exceeds real-time requirements, allowing for live integration into endoscopy suites.

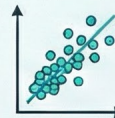
Clinical Performance & Reliability



0.8902 Mean Dice Coefficient
Demonstrates high spatial overlap and accuracy between AI predictions and expert ground-truth annotations.



0.9722 AUC-ROC Score
Indicates exceptional discriminative ability in distinguishing between polyps and healthy gastrointestinal tissue.



0.9364 Pearson Correlation
Shows a strong linear agreement between AI-calculated and manual expert measurements.

Comparison Against Baseline & SOTA Models (Kvasir-SEG)

Method	Dice Coefficient	IoU (Jaccard)	
ResUNet (Baseline)	0.7878	0.7778	
PolypSeg-GradCAM (Ours)	0.8902	0.8023	✓
PraNet	0.9053	0.8417	

Conclusion & Future Directions

Summary of Contributions



Balanced Accuracy & Trust.
(0.8902 Dice + Grad-CAM)



Rigorous Validation.
(5-Fold Cross-Validation on 1,000 images)



Deployment Ready.
(176 FPS real-time speed)

Future Roadmap



Multi-Center Validation.
(Testing on CVC-ClinicDB, ETIS datasets)



Video Sequence Testing.
(Evaluating temporal consistency)



Transformer Hybrids.
(Exploring lightweight Vision Transformers for edge cases)

PolypSeg-GradCAM successfully bridges the gap between Deep Learning performance and the clinical need for interpretable, trustworthy AI.

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Thank you for listening, Questions?

Contact:

- Ulas Bagci, Ph.D.,
- Professor, Director of Machine & Hybrid Intelligence Lab, Departments of Radiology, ECE, and BME
- Northwestern University, Chicago, IL 60611, USA
- Ulas.Bagci@northwestern.edu
- www.bagcilab.com

